Lecture 10. Hypothesis Testing (Part 2)

Trend Tests, Stratified Tests, Matched Pairs, Renyi-Type Tests

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1 Review: The K-Sample Test

Right-censored data $\{(T_{ik}, \delta_{ik})\}$; pooled death times $t_1 < \cdots < t_D$; d_j, d_{jk} deaths and Y_j, Y_{jk} at risk at t_j (pooled / group k). With a common weight W,

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\}, \qquad \chi^2 = (Z_1, \ldots, Z_{K-1})\,\hat{\Sigma}_{K-1}^{-1}\,(Z_1, \ldots, Z_{K-1})^T \overset{H_0}{\approx} \chi^2_{K-1},$$

with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1} d_j, \qquad \hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1} d_j.$$

This omnibus test is **not sensitive to a trend alternative**.

2 Tests for Trend

2.1 Order-specific alternatives

$$H_0 : h_1(t) = \cdots = h_K(t) \;\; \forall t \le \tau \qquad \text{vs.} \qquad H_A : h_1(t) \le h_2(t) \le \cdots \le h_K(t) \;\; \forall t \le \tau,$$

with at least one strict inequality; equivalently $H_A : S_1(t) \ge S_2(t) \ge \cdots \ge S_K(t)$.

2.2 Trend test statistic

Objective: increase power by a statistic sensitive to the stochastic ordering.

Definition: Trend test

For scores $a_1 < a_2 < \cdots < a_K$,

$$Z = \frac{\sum_{k=1}^{K} a_k Z_k(\tau)}{\left(\sum_{k=1}^{K}\sum_{k'=1}^{K} a_k a_{k'} \hat{\sigma}_{kk'}\right)^{1/2}} = \frac{\mathbf{a}'\mathbf{Z}}{\sqrt{\mathbf{a}'\hat{\Sigma}\mathbf{a}}},$$

with $\hat{\sigma}_{kk'}$ from the full $K \times K$ $\hat{\Sigma}$. Under H_0 each $Z_k(\tau)$ is a mean-zero martingale, so the linear combination is approximately normal with mean 0, and $Z \approx N(0,1)$. Reject H_0 in favour of H_A at level α if Z exceeds the upper α quantile of $N(0,1)$ (one-sided).

- **Choice of a_k**: heuristically, under H_A, $\mathrm{E}\{Z_1(\tau)\} \le \mathrm{E}\{Z_2(\tau)\} \le \cdots \le \mathrm{E}\{Z_K(\tau)\}$ (higher hazard $\Rightarrow$ more observed than expected deaths). Taking $a_k = k$ (standard) gives large weight to the groups with large expected values under H_A. Other choices may have better properties.
- The full $\hat{\Sigma}$ is singular, but $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ as long as $\mathbf{a}$ is not constant, so no group needs to be dropped.

2.3 Invariance

Theorem: Invariance under linear transformations of the scores

Let $Z_{\mathbf{a}} = \mathbf{a}'\mathbf{Z}/\sqrt{V[\mathbf{a}'\mathbf{Z}]}$ and $\mathbf{b} = c\mathbf{a} + d\mathbf{1}_K$ with constants $c > 0$, d. Then $Z_{\mathbf{b}}(\tau) = Z_{\mathbf{a}}(\tau)$.

Proof. Since $\sum_k Z_k(\tau) = 0$, i.e. $\mathbf{1}'\mathbf{Z} = 0$ (and hence $\hat{\Sigma}\mathbf{1} = \mathbf{0}$),

$$\mathbf{b}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z} + d\,\mathbf{1}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z}, \qquad V[\mathbf{b}'\mathbf{Z}] = V[c\,\mathbf{a}'\mathbf{Z}] = c^2 V[\mathbf{a}'\mathbf{Z}].$$

So $Z_{\mathbf{b}} = c\,\mathbf{a}'\mathbf{Z}/(c\sqrt{V[\mathbf{a}'\mathbf{Z}]}) = Z_{\mathbf{a}}$. (If $c < 0$ the sign flips.) ■

E.g. scores $(1,2,3,4)$ and $(0,1,2,3)$ or $(-3,-1,1,3)$ give the same test.

Example: Example 7.6 — larynx cancer by stage

90 larynx cancer patients in stages I–IV. KM curves (K&M Fig. 7.3) are ordered: stage I best, stage IV worst. Give the most severe disease the heaviest weight: $a_k = k$, $k = 1, \ldots, 4$. With log-rank weights,

$$\mathbf{Z}(10.7) = (-7.5660,\ -3.0117,\ 2.9155,\ 7.6623),$$

$$\hat{\Sigma} = \begin{pmatrix} 12.0740 & -4.4516 & -6.2465 & -1.3759 \\ -4.4516 & 7.8730 & -2.7599 & -0.6614 \\ -6.2465 & -2.7599 & 9.9302 & -0.9238 \\ -1.3759 & -0.6614 & -0.9238 & 2.9612 \end{pmatrix}.$$

Then $\mathbf{a}'\mathbf{Z} = -7.566 - 6.023 + 8.747 + 30.649 = 25.806$ and $\mathbf{a}'\hat{\Sigma}\mathbf{a} = 48.152$, so

$$Z = \frac{25.806}{\sqrt{48.152}} = 3.72, \qquad Z^2 = 13.83, \qquad p < 0.0001.$$

(The value 13.83 reported for (7.4.2) is Z^2.) With other weights: Tarone–Ware $Z = 4.06$, Gehan 4.22, Peto–Peto 4.13. All have $p < 0.0001$, strong evidence that the hazard rates are in the expected order.

Remark

Use a trend test only with strong prior information about the group ordering; otherwise it can lose power relative to the omnibus test.

3 Stratified Tests

3.1 Rationale

When comparing K groups we often need to adjust for other covariates. E.g. compare 3 vs. 6 months of chemotherapy in IBCSG adjusting for tumour stage (I–IV): the interest is the treatment, not the stage. One option is a regression model for covariate effects, but model-based adjustment is sensitive to misspecification. A conservative alternative is **stratification**.

3.2 Set-up

Definition: Stratified K-sample test

$H_0 : h_{1s}(t) = h_{2s}(t) = \cdots = h_{Ks}(t)$, $t \in [0, \tau]$, for each stratum $s = 1, \ldots, M$.

1. Using only the data in stratum s, compute $Z_{ks}(\tau)$, $k = 1, \ldots, K$, and $\hat{\Sigma}_s = \widehat{\mathrm{Cov}}[Z_{1s}(\tau), \ldots, Z_{Ks}(\tau)]$.
2. Global test: $Z_{k\cdot}(\tau) = \sum_{s=1}^{M} Z_{ks}(\tau)$ and $\hat{\Sigma}_{\cdot} = \widehat{\mathrm{Cov}}[Z_{1\cdot}, \ldots, Z_{K\cdot}] = \sum_{s=1}^{M} \hat{\Sigma}_s$ (strata are independent).
3. $\chi^2 = [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]^T \overset{H_0}{\sim} \chi^2_{K-1}$.

Within each stratum, observed and expected deaths are compared only among subjects with the same stratum, so differences in the baseline hazard across strata cannot masquerade as group effects.

Example: Example 7.4 — bone marrow transplant, stratified by MTX

The three disease groups (ALL, AML low risk, AML high risk) were compared in Lecture 9 and all tests rejected H_0. A more careful comparison accounts for whether patients received the graft-versus-host prophylactic methotrexate (MTX) or not (NOMTX). Stratified log-rank ($W = 1$):

$$\text{MTX:} \quad (Z_1, Z_2, Z_3) = (1.2816,\ -2.4929,\ 1.2113),$$

$$\hat{\Sigma}_{MTX} = \begin{pmatrix} 6.23336 & -3.42915 & -2.80422 \\ -3.42915 & 5.60732 & -2.17818 \\ -2.80422 & -2.17818 & 4.98239 \end{pmatrix},$$

$$\text{NOMTX:} \quad (Z_1, Z_2, Z_3) = (-0.504,\ -11.235,\ 11.739),$$

$$\hat{\Sigma}_{NOMTX} = \begin{pmatrix} 9.6771 & -6.5535 & -3.1236 \\ -6.5535 & 13.9816 & -7.4281 \\ -3.1236 & -7.4281 & 10.5517 \end{pmatrix}.$$

Summing: $Z_{1\cdot} = 0.7776$, $Z_{2\cdot} = -13.7279$, $Z_{3\cdot} = 12.9503$ (again linearly dependent), and

$$\hat{\Sigma}_{\cdot} = \begin{pmatrix} 15.91046 & -9.98265 & -5.92782 \\ -9.98265 & 19.58892 & -9.60628 \\ -5.92782 & -9.60628 & 15.53409 \end{pmatrix}.$$

Using the first two:

$$\chi^2 = (0.7776,\ -13.7279)\begin{pmatrix} 15.9105 & -9.9827 \\ -9.9827 & 19.5889 \end{pmatrix}^{-1}\begin{pmatrix} 0.7776 \\ -13.7279 \end{pmatrix} = 13.19 \quad (2 \text{ df},\ p \approx 0.001).$$

Remark: Remarks

Stratified tests have good power against alternatives in the *same direction* in every stratum. Otherwise, Z from one stratum can be cancelled by Z from another, the overall statistic may be near 0, and the test has very low power. In that case, run separate K-sample tests in each stratum.

4 Matched Pairs

Another use of stratification: each matched pair is a stratum. Paired times (T_{1i}, T_{2i}) and indicators $(\delta_{1i}, \delta_{2i})$, $i = 1, \ldots, M$; test $H_0 : h_{1i}(t) = h_{2i}(t)$. With $K = 2$, in each pair

$$Z_{1i}(\tau) = \sum_j W(t_j)\Big\{d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big\}, \qquad \sigma_{11i} = \sum_j W(t_j)^2 \frac{Y_{j1}}{Y_j}\Big(1 - \frac{Y_{j1}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}d_j.$$

Derivation: within a single pair

Before the first time in the pair both are at risk: $Y_j = 2$, $Y_{j1} = 1$, so $Y_{j1}/Y_j = 1/2$.

1. $T_{1i} = T_{2i}$: $d_j = \delta_{1i} + \delta_{2i}$,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i} + \delta_{2i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i} - \delta_{2i}}{2},$$

$$\sigma_{11i} = W^2(T_{1i})\tfrac{1}{2}(1 - \tfrac{1}{2})\frac{2 - \delta_{1i} - \delta_{2i}}{2 - 1}(\delta_{1i} + \delta_{2i})$$

$$= \frac{W^2(T_{1i})}{4}(2 - \delta_{1i} - \delta_{2i})(\delta_{1i} + \delta_{2i}).$$

2. $T_{1i} < T_{2i}$: only the first time can be an event with both at risk,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i}}{2}, \qquad \sigma_{11i} = \frac{W^2(T_{1i})}{4}(2 - \delta_{1i})\delta_{1i}.$$

3. $T_{1i} > T_{2i}$: $Z_{1i} = -W(T_{2i})\dfrac{\delta_{2i}}{2}$, $\quad \sigma_{11i} = \dfrac{W^2(T_{2i})}{4}(2 - \delta_{2i})\delta_{2i}$.

(After the first time, one of the pair is gone, so $Y_{j1}/Y_j \in \{0, 1\}$ and later terms contribute nothing.) For 0/1 indicators $(2 - \delta)\delta = \delta$, so:

$$Z_{1i} = \begin{cases} W(T_{1i})/2, & T_{1i} < T_{2i}, \delta_{1i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{1i} = 1, \delta_{2i} = 0, \\ -W(T_{2i})/2, & T_{2i} < T_{1i}, \delta_{2i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{2i} = 1, \delta_{1i} = 0, \\ 0, & \text{otherwise;} \end{cases}$$

$$\hat{\sigma}_{11i} = \begin{cases} W(T_{1i})^2/4, & \text{first case,} \\ W(T_{2i})^2/4, & \text{second case,} \\ 0, & \text{otherwise.} \end{cases}$$

Theorem: Paired log-rank (sign-type) test

If W does not depend on time (e.g. log-rank, $W \equiv 1$), with $D_1 = \#$ pairs in which member 1 has the event first and $D_2 = \#$ pairs in which member 2 does,

$$Z_{1\cdot} = \frac{W}{2}(D_1 - D_2), \qquad \hat{\sigma}_{11\cdot} = \sum_{i=1}^{M} \hat{\sigma}_{11i} = \frac{W^2}{4}(D_1 + D_2), \qquad \frac{Z_{1\cdot}}{\sqrt{\hat{\sigma}_{11\cdot}}} = \frac{D_1 - D_2}{\sqrt{D_1 + D_2}} \overset{H_0}{\approx} N(0,1).$$

This is the sign test (McNemar-type): under H_0 each informative pair is equally likely to go either way.

Example: Example 7.8 — 6-MP vs. placebo matched pairs

$D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$, so $Z = (18-3)/\sqrt{18+3} = 3.27$ and $p = 2P[Z > 3.27] = 0.001$: sufficient evidence that the relapse rates differ.

5 Renyi-Type Tests

5.1 Rationale

If the hazards of the two groups **cross**, the usual two-sample tests may have low power: early differences cancel late differences of opposite sign. Renyi-type tests are built on the *absolute values of cumulative differences*; a Renyi statistic is the analogue for censored data of the Kolmogorov–Smirnov statistic.

5.2 Set-up and statistic

Test $H_0 : h_1(t) = h_2(t)$, $t \le \tau$, vs. $H_A : h_1(t) \neq h_2(t)$ for some $t \le \tau$. Two samples of sizes n_1, n_2; pooled distinct failure times $t_1 < \cdots < t_D$; d_{j1}, d_{j2} and Y_{j1}, Y_{j2} with $d_j = d_{j1} + d_{j2}$, $Y_j = Y_{j1} + Y_{j2}$.

Definition: Renyi statistic

Cumulative weighted difference up to t_i:

$$Z(t_i) = \sum_{t_j \le t_i} W(t_j)\Big[d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big], \qquad i = 1, \ldots, D,$$

with standard error at τ (largest t_j with $Y_{j1}, Y_{j2} > 0$)

$$\sigma^2(\tau) = \sum_{t_j \le \tau} W^2(t_j)\frac{Y_{j1}}{Y_j}\frac{Y_{j2}}{Y_j}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

The two-sided test statistic is

$$Q = \frac{\sup\{|Z(t)|,\ t \le \tau\}}{\sigma(\tau)}.$$

Theorem: Null distribution

Under H_0, Q is approximately distributed as $\sup\{|B(x)|,\ 0 \le x \le 1\}$, B a standard Brownian motion. Critical values are in K&M Table C.5.

Remark: Why Brownian motion

$Z(t)$ is a martingale in t with variance process $\sigma^2(t)$; rescaling time by $x = \sigma^2(t)/\sigma^2(\tau) \in [0,1]$, $Z/\sigma(\tau)$ behaves like $B(x)$. The supremum of $|Z|$ catches a large excursion even if $Z(\tau)$ returns near 0.

Comparison with the usual test. The usual statistic is $Z(\tau)/\sqrt{\text{Var}[Z(\tau)]}$ (or its square). When hazards cross, early positive differences are cancelled by later negative ones, and the log-rank test may lack power; the supremum test based on $|Z|$ has greater power for such differences.

Example: Example 7.9 — gastric cancer: chemotherapy vs. chemotherapy + radiotherapy

The KM curves cross (K&M Fig. 7.5): chemo + radiation is worse early but better later. Using the log-rank version of the Renyi statistic, $|Z(t_j)|$ (Fig. 7.4) peaks at $t_j = 315$ with value 9.8; $\sigma(\tau = 2363) = 4.46$, so

$$Q = 9.8/4.46 = 2.2, \qquad 0.05 < p < 0.06 \text{ (Table C.5)},$$

close to rejecting H_0. The usual log-rank test gives $Z(2363) = -2.15$, statistic $-2.15/4.46 = -0.482$, $p = 2P(Z < -0.482) = 0.63$: it misses the crossing difference entirely.